ARABINDA BEHERA

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PROFESSIONAL SUMMARY

- Internationally recognized scientist and software engineer with original contributions spanning experimental high-energy nuclear physics and enterprise AI systems. Graduated with a Ph.D degree in Chemistry from Stony Brook University with research conducted at CERN (ATLAS Collaboration) and Brookhaven National Laboratory (STAR Collaboration), and currently serves in a critical engineering role at Oracle America, Inc.
- Qualified through sustained scientific contributions to be included in the official author lists of the ATLAS Collaboration at CERN and the STAR Collaboration at BNL — a status granted only to scientists who meet rigorous service, analysis, and authorship requirements set by these collaborations of 3,000+ and 600+ international researchers respectively.
- Made original methodological and computational contributions to large-scale experimental data analysis, including advancing statistical frameworks for collective flow extraction and pioneering deep generative model applications for high-fidelity detector simulation.
- Authored 20+ peer-reviewed publications across the most selective venues in physics — including *Nature*, *Nature Physics*, and *Science Advances*, alongside Physical Review Letters, Physical Review C, Physical Review Research, JHEP, Physics Letters B, and Computing and Software for Big Science — cited several hundred times by independent researchers worldwide.
- Named co-author on landmark discoveries published in *Nature*: the observation of quantum entanglement with top quarks at the LHC (2024); the detailed ten-year map of Higgs boson interactions (2022); global spin alignment of ϕ and K^{*0} vector mesons in heavy-ion collisions (2023); and global Λ-hyperon polarization establishing the quark-gluon plasma as the most vortical fluid ever observed (2017) — each the subject of worldwide press coverage (Nature News, CERN, Brookhaven National Laboratory, U.S. DOE, NSF, Physics World, BBC) and contributing to the body of work recognized by the 2025 Breakthrough Prize in Fundamental Physics.
- Invited speaker and published proceedings author at premier international physics conferences.
- 5+ years experience Software Development. Designed and delivered cloud-native full-stack applications and AI-powered systems at enterprise scale, spanning distributed microservices, generative AI agents, retrieval-augmented systems, and modern web platforms.
- Holds advanced industry credentials in Generative AI, Cloud Architecture, Data Science, and Machine Learning from Oracle, DeepLearning.AI, IBM, Google Cloud, and Cloudera — reflecting continued commitment to remaining at the frontier of the field.
- Operating at the rare intersection of fundamental scientific research and large-scale production engineering — applying statistical modeling, high-performance computing, Monte Carlo simulation, deep learning, and distributed-systems expertise to problems that few practitioners worldwide can address with equivalent depth.

EDUCATION

- **State University of New York at Stony Brook** — Stony Brook, USA
  Ph.D in Chemistry — Aug 2015 - May 2021
- **National Institute of Science Education and Research** — Bhubaneswar, India
  Int. MSc in Physics — Aug 2010 - May 2015

SKILLS

Programming	:	Java, Javascript, C, C++, Python, Shell, HTML, PL/SQL
Frameworks	:	Spring Boot, React, NextJS, Helidon, Express, FastAPI, Flask
Database	:	Oracle Database, Postgresql, MySQL, MongoDB, Redis, Apache Hive
Cloud	:	OCI, AWS, GCP, Azure
Devops	:	Jenkins, Docker, Kubernetes, Terraform, Ansible
ML Libs	:	Scikit-Learn, Tensorflow, Keras, Jupyter, Numpy, Pandas, Matplotlib, Seaborn
AI tools	:	OpenAI, Google Gemini, Claude Code, RAG, LangChain, LangGraph
Others	:	JUnit, Mockito, Kafka, Git, JIRA, Bitbucket, Confluence, Linux

EXPERIENCE

- **Oracle America, Inc.** — Feb 2022 - Present
  Software Engineer — *Redwood City, CA*
  - Designed and developed cloud-native full-stack applications for Oracle Fusion Release Engineering platform, supporting large-scale software delivery workflows across Oracle's enterprise product suite.
  - Built production microservices using Java and the Helidon framework, integrated with Oracle Autonomous Database and OCI cloud services to handle critical release pipeline operations.
  - Implemented CI/CD pipelines and automated deployment workflows using Oracle Cloud Infrastructure (OCI), reducing manual release overhead and increasing deployment frequency.
  - Designed and developed REST APIs with comprehensive unit and integration testing using JUnit and Mockito, ensuring high reliability and adherence to engineering standards.
  - Built and deployed internal web applications using the Oracle APEX low-code platform and frontend applications using JavaScript, HTML, CSS, and Oracle JET.
  - Designed and developed a CLI tool in Golang and OCI Cloud services to automate the export, import, authorization, and deployment of Oracle APEX applications, significantly reducing manual effort for developer teams.
  - Designed and implemented a Retrieval Augmented Generation (RAG) system using Oracle Autonomous Vector Database, OCI services like Knowledge base and embedding models, enabling semantic search over developer logs and reducing debugging turnaround time.
  - Developed an AI Chat Agent using Oracle Generative AI service backed by a structured knowledge base of technical documentation, improving developer self-service and reducing support load.
- **Revize Software Systems** — Aug 2021 - Feb 2022
  Software Developer — *Troy, MI*
  - Maintained and extended the Java-based Revize CMS platform serving government institutions — municipalities, public agencies, and local governments across the United States — implementing new features, resolving production bugs, and improving platform stability using Spring Boot, Hibernate, and PostgreSQL; collaborated via Git, JIRA, and Confluence in an Agile sprint workflow.
  - Decomposed monolithic CMS modules into independently deployable Spring Boot microservices, improving platform scalability and reducing inter-component coupling; wrote comprehensive unit and integration tests using JUnit and Mockito to enforce reliability across service boundaries.
  - Developed dynamic, client-facing frontend pages and UI components for government websites using PHP, JavaScript, HTML/CSS, and jQuery, delivering responsive layouts and interactive features directly consumed by thousands of public-sector end users.
  - Architected and delivered a full-stack accident reporting SPA for police department use: React frontend, Spring Boot REST backend, MySQL database — deployed end-to-end on AWS with EC2 instances, Application Load Balancers, AWS WAF rules, and IAM security policies, replacing a manual paper-based workflow with a secure, cloud-hosted digital system.

- **Resideo(Honeywell Home)** June 2021 - Aug 2021
  Firmware Engineer Intern *Melville, NY*
  - Implemented Apple HomeKit Accessory Protocol (HAP) over IP for residential security systems (alarm panels, door locks, motion sensors), enabling Siri voice control and Apple Home compatibility; built server/client processes on embedded Linux using socket-based IPC and event-driven architecture for real-time device state synchronization between the HomeKit bridge daemon and security panel firmware.
  - Designed and developed a cross-platform C++ SDK abstracting low-level hardware communication protocols for programmatic control of Resideo security panels and peripheral IoT devices, with TLS-based secure channels to the Resideo cloud backend ensuring Apple MFi certification compliance.
  - Conducted functional and integration testing of HomeKit-paired security devices using Apple's HomeKit Accessory Simulator and hardware test rigs, validating protocol compliance and device state transitions across multiple firmware versions. feature delivery for the HomeKit integration track.
- **SUNY at Stony Brook** June 2016 - May 2021
  Research Assistant *Stony Brook, NY*
  - Performed large-scale data analysis of nuclear collision experiments for the ATLAS experiment at CERN and the STAR experiment at BNL, contributing to globally significant particle physics research.
  - Designed and optimized High-Level Trigger (HLT) algorithms for electron and photon selection in the ATLAS experiment at the LHC, achieving trigger efficiencies exceeding 96% for pp collisions and 95% for heavy-ion collisions while adapting to a fourfold increase in peak LHC luminosity across the full Run 2 period (2015–2018); participated in live data-taking operations across pp, p+Pb, and Pb+Pb collision runs, validating trigger performance on raw data under rapidly increasing pile-up conditions, with contributions directly enabling the ATLAS physics program spanning Standard Model measurements and searches for new phenomena. (Eur. Phys. J. C 80, 47 (2020))
  - Developed the theoretical Glauber model framework linking longitudinal eccentricity decorrelations to measurable flow decorrelation coefficients across multiple heavy-ion collision systems, then co-authored the first experimental measurement of longitudinal flow decorrelations in Xe+Xe collisions at $\sqrt{s_{NN}} = 5.44$ TeV with the ATLAS detector — uncovering a unique system-size reversal between v_2 and v_3 decorrelations not present in inclusive flow harmonics, and demonstrating that current hydrodynamic models fail to describe the three-dimensional longitudinal structure of the QGP initial state; results were personally presented at four major international conferences including Quark Matter 2019 (Wuhan), Hard Probes 2020 (Texas), and Initial Stages 2021, with published proceedings. (Phys. Rev. Research 2, 023362 (2020); Phys. Rev. Lett. 126, 122301 (2021); PoS(HardProbes2020)117)
  - Performed critical evaluation of the robustness of Principal Component Analysis (PCA) as applied to harmonic flow extraction in heavy-ion collisions, using AMPT and HIJING Monte Carlo simulations. Demonstrated sensitivity of extracted leading and subleading flow modes to particle weight choice, transverse momentum range, and non-flow contamination from dijets — establishing fundamental methodological boundaries that directly inform how the global heavy-ion physics community interprets QGP collective behavior measurements at RHIC and the LHC. (Phys. Rev. C 102 (2020), 024911).
  - Developed and validated novel deep generative models — Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) — for fast photon shower simulation in the ATLAS electromagnetic calorimeter, achieving up to two orders of magnitude speedup over full GEANT4 simulation while accurately reproducing shower energy distributions across particle energies from 1 to 262 GeV on real ATLAS detector geometry; among the first such applications on actual detector readout rather than simplified calorimeter models. This work directly addressed the large-scale Monte Carlo production bottleneck critical to the LHC physics program and demonstrated the feasibility of deep learning as a scalable alternative for future ATLAS fast simulation infrastructure. (Comput Softw Big Sci 8, 7 (2024))
  - Earned named authorship — through ATLAS's and STAR's contribution-gated process (service, anal-

ysis, and peer-review requirements) — on four landmark results published in *Nature*, each generating global press coverage (Nature News, CERN, Brookhaven National Laboratory, U.S. DOE, NSF, Physics World, BBC): observation of quantum entanglement with top quarks (*Nature* 633, 2024); ten-year map of Higgs boson interactions (*Nature* 607, 2022); global spin alignment of ϕ and K^{*0} mesons (*Nature* 614, 2023); and global Λ-hyperon polarization — the most vortical fluid ever observed (*Nature* 548, 2017).
  - Published 20+ peer-reviewed papers — including four in *Nature* — across PRL, PRC, PRR, JHEP, PLB, *Nature*, *Nature Physics*, and *Science Advances*, cited hundreds of times internationally.

- **SUNY at Stony Brook** — Aug 2015 - June 2016
  Teaching Assistant — *Stony Brook, NY*
  - Assisted in undergraduate chemistry courses, conducted tutorial sessions, graded assignments, and proctored examinations for a class of 100+ students.

CERTIFICATIONS

- OCI Generative AI Professional (Oracle link)
- OCI Solution Architect Associate (Oracle link)
- Oracle Machine Learning using Autonomous Database Associate (Oracle link)
- OCI Data Science Professional (Oracle link)
- DeepLearning.AI Deep Learning Specialization (Coursera link)
- IBM Data Science Spcialization (Coursera link)
- Google Cloud Training Machine Learning Specialization (Coursera link)
- Cloudera Modern Big Data Analysis with SQL Specialization (Coursera link)

KEY PROJECTS

- **Job Scheduling and Monitoring Service -**
  - Developed microservices using Spring Boot for scheduling jobs and monitoring job status
  - Implemented authentication and authorization for job requests and integrated with OCI services like IAM, Compute, Object Storage and Autonomous Database for scheduling jobs.
  - Designed and developed a real time monitoring system for scheduled jobs using Apache Kafka and OCI Streaming service.
- **Developer Productivity Tool -**
  - Developed CLI using Golang for developer productivity
  - Automated creation, export, import, authorization and deployment of Oracle APEX applications
  - Reduced manual testing and deployment effort
- **AI RAG Log Search System -**
  - Built semantic search system using vector embeddings
  - Used Oracle Autonomous Vector Database
  - Implemented similarity search on developer logs
  - Improved debugging and log search efficiency
- **AI Facial Emotion Detection -**
  - Trained a DNN model on face image dataset to identify the key facial points which are the x- and y-coordinates of points capturing the positions of eyes, nose and lips.

- The model contained Convolutional layers, Residual blocks, Max-pooling layers and Dense layers and was trained with Adam optimization and Dropout.

AWARDS

- Breakthrough Prize in Fundamental Physics (link) - ATLAS, CERN (2025) - one of the most prestigious prizes in physics.
- Best Master's Thesis Award in Physics - NISER (2015)
- INSPIRE Fellowship Recipient (2010-2015) - Competitive national fellowship for top physics students in India.
- Chief Minister's Award for Academic Excellence (2010) - State-level recognition for outstanding academic achievement.

CONFERENCES

1. **Initial Stages 2021 Online** Rehovot, Israel, 2021
   - Title: *"Correlations between flow and transverse momentum in Pb+Pb and Xe+Xe collisions with ATLAS"*, Link - https://indico.cern.ch/event/854124/contributions/4134918/
2. **APS April Meeting 2021** U.S., 2021
   - Talk Title: *"Correlations of flow and transverse momentum in Pb+Pb and Xe+Xe collisions with ATLAS"*, Link - https://meetings.aps.org/Meeting/APR21/Session/K14.4
3. **Hard Probes 2020** Texas, U.S., 2020
   - Talk Title: *"FATLAS measurements of transverse and longitudinal flow decorrelations in Xe+Xe, Pb+Pb, and p+Pb collisions"*, Link: https://indico.cern.ch/event/751767/contributions/3770971/
   - Poster Title: *"Longitudinal flow decorrelation in Xe+Xe and Pb+Pb collisions with the ATLAS detector"*, Link: https://indico.cern.ch/event/751767/contributions/3775957/
   - Proceedings: A. Behera, *"Measurement of transverse flow and longitudinal flow decorrelations in 5.44 TeV Xe+Xe collisions with ATLAS"*, PoS(HardProbes2020)117
4. **2020 Fall Meeting of the APS DNP** U.S., 2020
   - Talk Title: *"Flow and transverse momentum correlations in Pb+Pb and Xe+Xe collisions with ATLAS*, Link: https://meetings.aps.org/Meeting/DNP20/Session/EC.5
5. **Quark Matter 2019** Wuhan, China, 2019
   - Poster Title: *"Measurements of longitudinal flow decorrelations in Xe+Xe collisions with the ATLAS detector*, Link: https://indico.cern.ch/event/792436/contributions/3533944/
6. **Initial Stages 2019** New York, U.S., 2019
   - Talk Title: *"Flow Fluctuation and Centrality Fluctuation in Pb+Pb collisions with the ATLAS Detector"*
     Link: https://indico.bnl.gov/event/5391/contributions/29239/
7. **WPCF 2018** Krakow, Poland, 2018
   - Talk Title: *"Measurement of long-range azimuthal correlations in proton-proton and proton-lead collisions with ATLAS"*, Link: https://indico.ifj.edu.pl/event/199/contributions/1123/
   - Proceedings: A. Behera, *"Measurement of long-range azimuthal correlations in proton-proton and proton-lead collisions with ATLAS"*, Acta Physica Polonica B Proceedings Supplement Vol. 12 (2019)
8. **Quark Matter 2017** Chicago, U.S., 2017

- Poster Title: *"Studying collectivity in small collision systems with multi-particle azimuthal correlations with the ATLAS detector"*, Link: https://indico.cern.ch/event/433345/contributions/2358170/

SELECTED PUBLICATIONS

Google Scholar - https://scholar.google.com/citations?hl=en&user=o-wsejEAAAAJ
Inspire HEP - https://inspirehep.net/authors/1403410
ORCID - https://orcid.org/0000-0002-7739-295X

1. ATLAS Collaboration, *"Observation of quantum entanglement with top quarks at the ATLAS detector"*, Nature 633, 542–547 (2024)
2. ATLAS Collaboration, *"Deep Generative Models for Fast Photon Shower Simulation in ATLAS"*, Computing and Software for Big Science Vol 8 #7 (2024)
3. STAR Collaboration, *"Pattern of global spin alignment of ϕ and K^{*0} mesons in heavy-ion collisions"*, Nature 614, 244–248 (2023)
4. ATLAS Collaboration, *"Correlations between flow and transverse momentum in Xe+Xe and Pb+Pb collisions at the LHC with the ATLAS detector: A probe of the heavy-ion initial state and nuclear deformation"*, Phys. Rev. C 107 (2023), 054910
5. ATLAS Collaboration, *"Observation of electroweak production of two jets and a Z-boson pair"*, Nature Phys. 19 (2023) 237–253
6. STAR Collaboration, *"Tomography of ultrarelativistic nuclei with polarized photon-gluon collisions"*, Sci. Adv. 9 (2023) eabq3903
7. ATLAS Collaboration, *"A detailed map of Higgs boson interactions by the ATLAS experiment ten years after the discovery"*, Nature 607, 52–59 (2022)
8. ATLAS Collaboration, *"Longitudinal flow decorrelations in Xe+Xe collisions at 5.44 TeV with the ATLAS detector"*, Phys. Rev. Lett. 126 (2021) 122301
9. C. Zhang, A. Behera, S. Bhatta, J. Jia, *"Non-flow effects in correlation between harmonic flow and transverse momentum in nuclear collisions"*, Phys.Lett.B 822 (2021) 136702
10. ATLAS Collaboration, *"Test of the universality of τ and μ lepton couplings in W-boson decays with the ATLAS detector"*, Nature Phys. 17 (2021) 813–818
11. A. Behera, *"Measurement of transverse flow and longitudinal flow decorrelations in 5.44 TeV Xe+Xe collisions with ATLAS"*, PoS(HardProbes2020)117
12. A. Behera, M. Nie, J. Jia, *"Longitudinal eccentricity decorrelations in heavy ion collisions"*, Phys. Rev. Research 2 (2020), 023362
13. Z. Liu, A. Behera, H. Song, J. Jia, *"Robustness of principal component analysis on harmonic flow in heavy ion collisions"*, Phys. Rev. C 102 (2020), 024911
14. ATLAS Collaboration, *"Performance of electron and photon triggers in ATLAS during LHC Run 2"*, Eur. Phys. J. C 80 (2020) 47
15. STAR Collaboration, *"Measurement of the mass difference and the binding energy of the hypertriton and antihypertriton"*, Nature Phys. 16 (2020) 409–412
16. ATLAS Collaboration, *"Fluctuations of anisotropic flow in Pb+Pb collisions at $\sqrt{s_{NN}}$=5.02 TeV with the ATLAS detector"*, JHEP 01 (2020) 51
17. ATLAS Collaboration, *"Correlated long-range mixed-harmonic fluctuations measured in pp, p+Pb and low-multiplicity Pb+Pb collisions with the ATLAS detector"*, Phys. Lett. B 789 (2019) 444
18. A. Behera, *"Measurement of long-range azimuthal correlations in proton-proton and proton-lead collisions with ATLAS"*, Acta Physica Polonica B Proceedings Supplement Vol. 12 (2019)
19. ATLAS Collaboration, *"Measurement of flow harmonics correlations with mean transverse momentum in lead–lead and proton–lead collisions at $\sqrt{s_{\mathrm{NN}}} = 5.02$ TeV with the ATLAS detector"*, Eur. Phys. J. C 79 (2019) 985

20. ATLAS Collaboration, *"Correlated long-range mixed-harmonic fluctuations measured in pp, p+Pb and low-multiplicity Pb+Pb collisions with the ATLAS detector"*, Phys. Lett. B 789 (2019) 444

21. ATLAS Collaboration, *"Observation of Higgs boson production in association with a top quark pair at the LHC with the ATLAS detector"*, Phys. Lett. B 784 (2018) 173

22. STAR Collaboration, *"Global Λ hyperon polarization in nuclear collisions: evidence for the most vortical fluid"*, Nature 548, 62 (2017)

23. STAR Collaboration, *"Bulk Properties of the Medium Produced in Relativistic Heavy-Ion Collisions from the Beam Energy Scan Program"*, Phys. Rev. C 96, 044904 (2017)